Digital Avionics
航電系統

Lecture 4

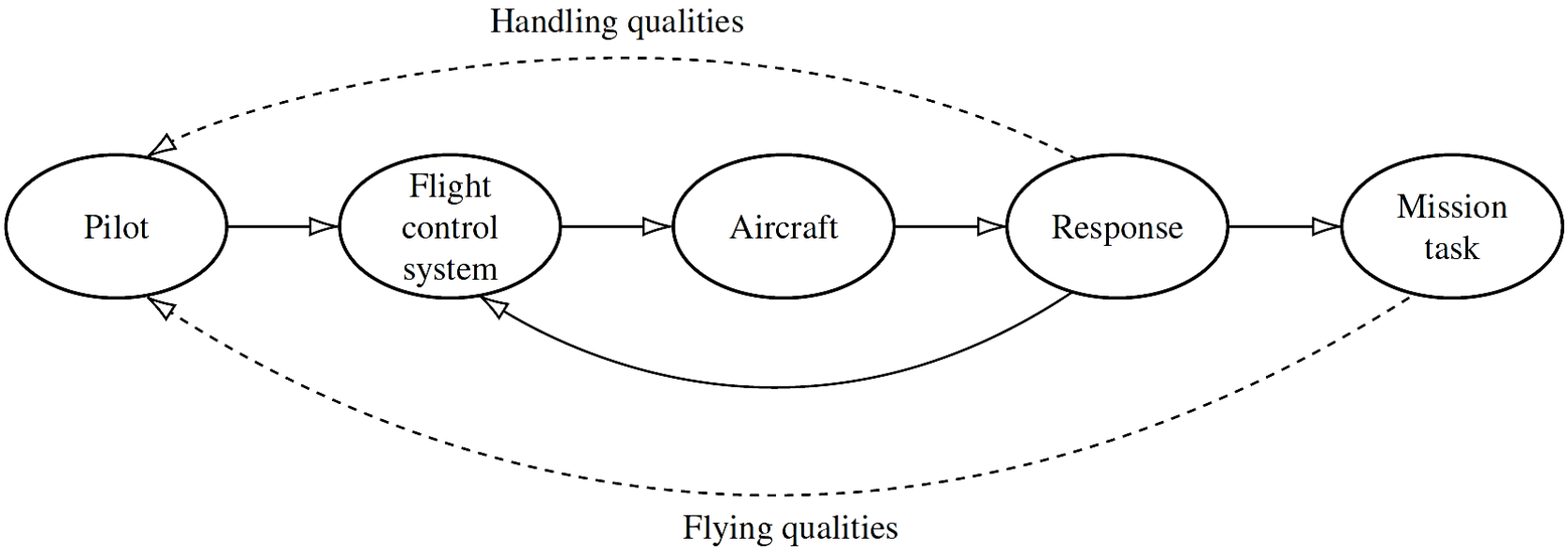
Prof. Ying-Chih Lai

Textbook

- Reading:
  - Chapter 3: Aerodynamics and Aircraft Control
  - Section 3.1~3.8, pp. 101-162
- 3.4 Aircraft Dynamics
- 3.5 Longitudinal Control and Response
- 3.8 Stability Augmentation Systems

Flying and Handling Quality

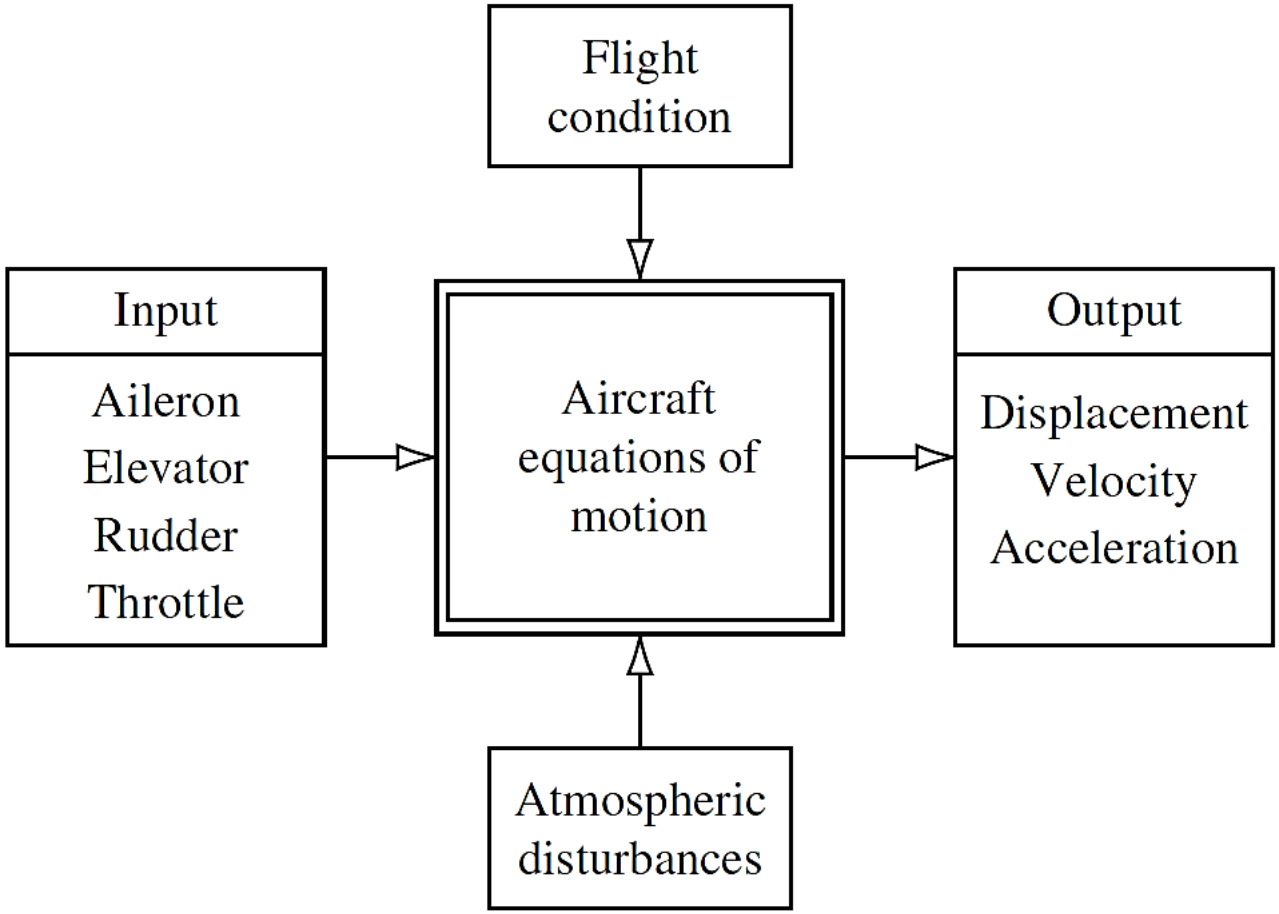
- Flying and handling qualities of FBW aircraft



The pilot has direct control only of the Flight Control System. However, he can tailor his inputs to the FCS by observing the airplane's response while always keeping an eye on the task at hand.

Mathematical Model

- Basic control-response relationships



Models of Dynamic Systems

- Mathematical Models of Dynamic Systems are Differential Equations

Continuous-time dynamic process: Vector Ordinary Differential Equation

$$\dot{\mathbf{x}}(t) \triangleq \frac{d\mathbf{x}(t)}{dt} = \mathbf{f}[\mathbf{x}(t), \mathbf{u}(t), \mathbf{w}(t), \mathbf{p}(t), t]$$

$$\dim(\mathbf{x}) = (n \times 1)$$
$$\dim(\mathbf{f}) = (n \times 1)$$
$$\dim(\mathbf{u}) = (m \times 1)$$
$$\dim(\mathbf{w}) = (s \times 1)$$
$$\dim(\mathbf{p}) = (l \times 1)$$

Output Transformation

$$\mathbf{y}(t) = \mathbf{h}[\mathbf{x}(t), \mathbf{u}(t)]$$

$$\dim(\mathbf{y}) = (r \times 1)$$
$$\dim(\mathbf{h}) = (r \times 1)$$

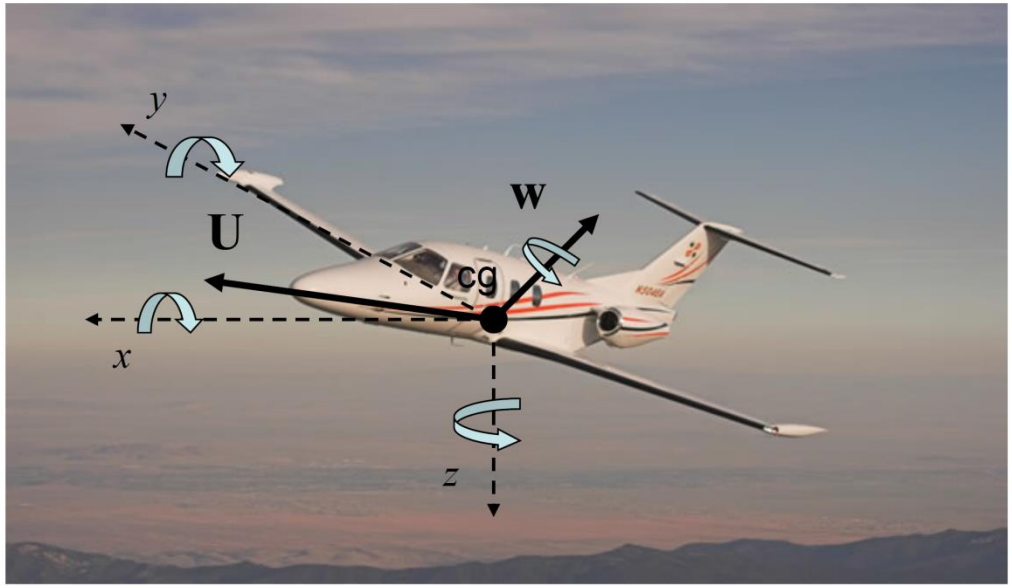
Measurement with Error

$$\mathbf{z}(t) = \mathbf{y}(t) + \mathbf{n}(t)$$

$$\dim(\mathbf{z}) = (r \times 1)$$
$$\dim(\mathbf{n}) = (r \times 1)$$

Aircraft Degrees of Freedom

- Six degrees of freedom (6 DoF):
  - 3 displacements
    - *x*: horizontal motion
    - *y*: side motion
    - *z*: vertical motion
  - 3 rotations
    - Around *x*: roll
    - Around *y*: pitch
    - Around *z*: yaw



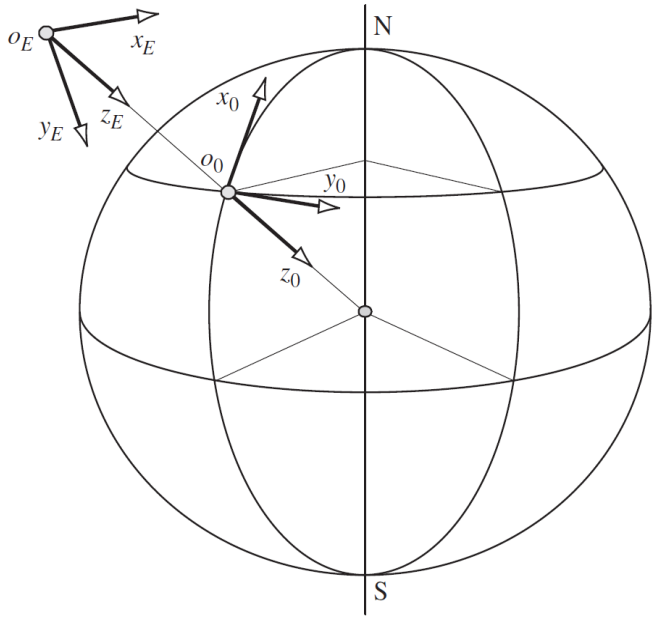
U: resultant linear velocity, cg: center of gravity
w: resultant angular velocity

Aircraft Frames of Reference

- There are many possible coordinate systems:
  - Inertial (immobile and far away)
  - Earth-fixed (rotates with the earth's surface)
  - Vehicle carried vertical frame (fixed on aircraft cg, vertical axis parallel to gravity)
  - Air-trajectory (fixed on aircraft cg, parallel to the direction of motion of the aircraft)
  - Body-fixed (fixed on aircraft cg, parallel to a geometric datum line on the aircraft)
  - Stability axes (fixed on aircraft cg, parallel to a reference flight condition)
  - Others

Systems of Axes and Notation

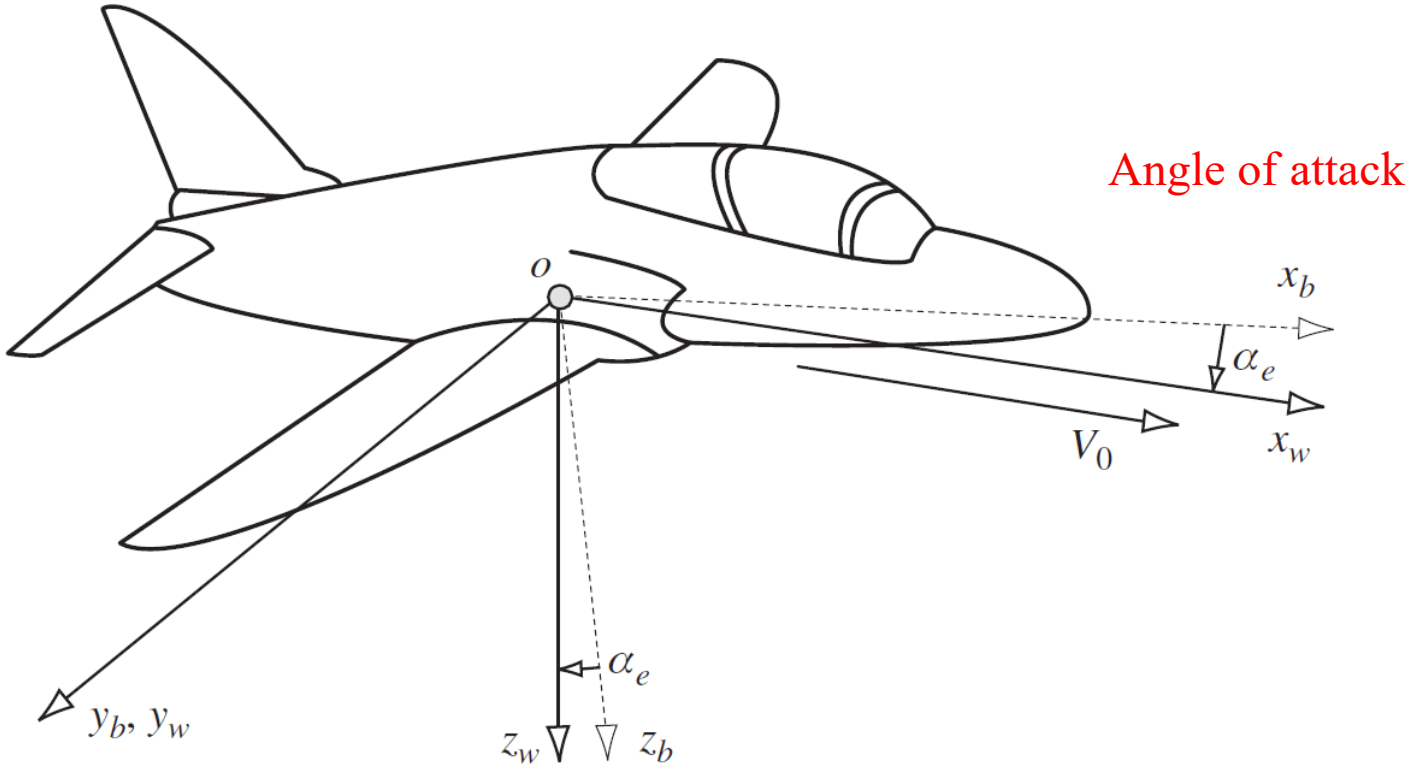
- Earth-fixed Axes
  - Since only short-term motion is of interest, it is perfectly adequate to assume flight above a flat earth.
  - The most common consideration is that of motion about *straight* and *level* flight.
  - The horizontal plane is defined by($o_E x_E y_E$) and is parallel to the plane ($o_0 x_0 y_0$) at the surface of the earth.



Conventional earth axes.

Moving-Axis (Body-Fixed) Systems

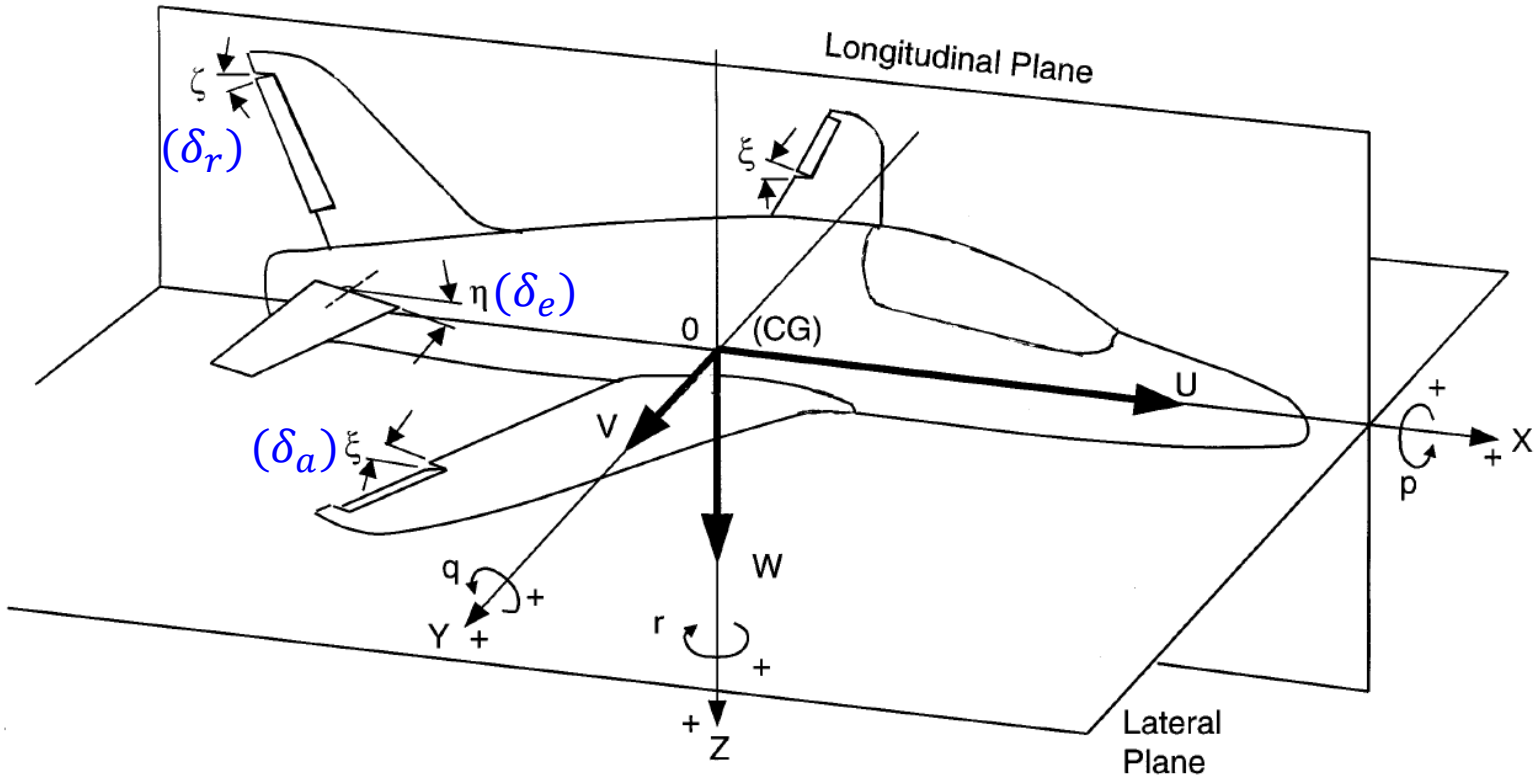
- Body axes rotated about the oy_b axis through the steady body incidence angle α_e until the ox_w axis aligns with the velocity vector.
- Typically, the body incidence might vary in the range $-10° < \alpha_e < 20°$ over a normal flight envelope.



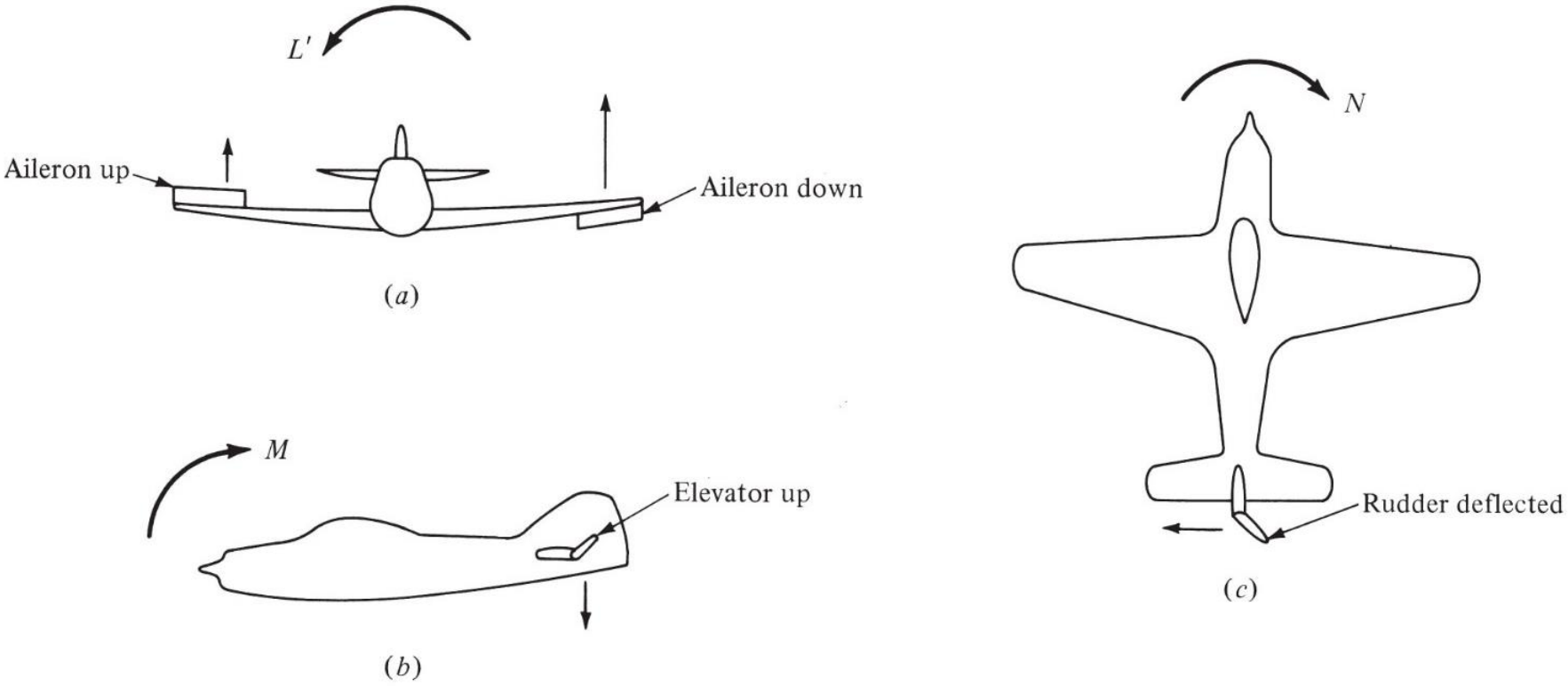
Controls Notation

- Body axis and motion in 3 axis coordinate system are defined as:

AXIS	Linear Velocity Component	Angular Velocity Component
Forward or Roll Axis, 0X	Forward Velocity U	Roll Rate p
Sideslip or Pitch Axis, 0Y	Sideslip Velocity V	Pitch Rate q
Vertical or Yaw Axis, 0Z	Vertical Velocity W	Yaw Rate r



Controls Notation



Effect of control deflections on roll, pitch, and yaw. (a) Effect of aileron deflection; lateral control, (b) Effect of elevator deflection; longitudinal control, (c) Effect of rudder deflection; directional control.

Controls Notation

- Control surface displacement

In roll: positive right push force on the stick ⇒ positive stick displacement ⇒ right aileron up and left aileron down (negative mean) ⇒ right wing down roll response (positive).
In pitch: positive pull force on the stick ⇒ positive aft stick displacement ⇒ elevator trailing edge up (negative) ⇒ nose-up pitch response (positive).
In yaw: positive push force on the right rudder pedal ⇒ positive rudder bar displacement ⇒ rudder trailing edge displaced to the right (negative) ⇒ nose to the right yaw response (positive).

- Engine control
  - Engine thrust τ is controlled by throttle lever displacement ε.

$$\frac{\tau(s)}{\varepsilon(s)} = \frac{k_\tau}{(1 + sT_\tau)} \qquad (2.34)$$

  - where k_τ is a suitable gain constant and T_τ is the lag time constant, which is typically of the order of two to three seconds.

Linear Quantities Transformation

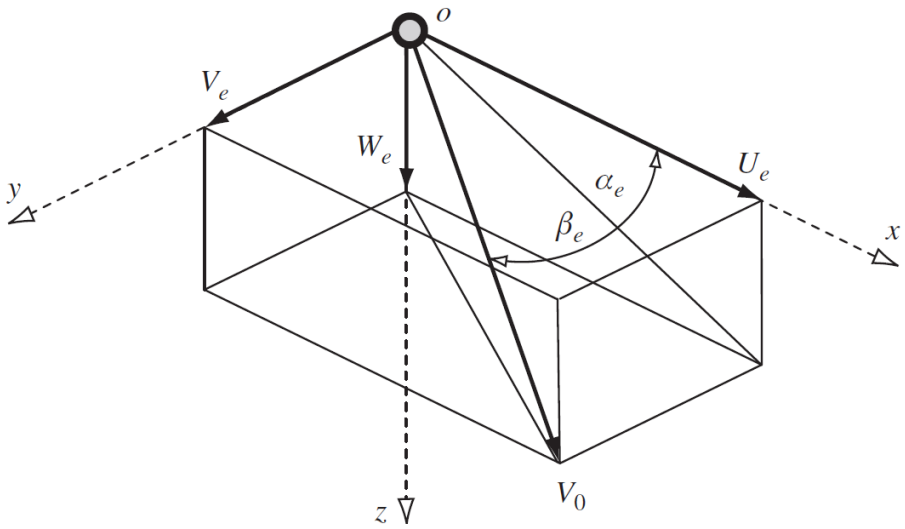


FIGURE 2.6 Resolution of velocity through incidence and sideslip angles.

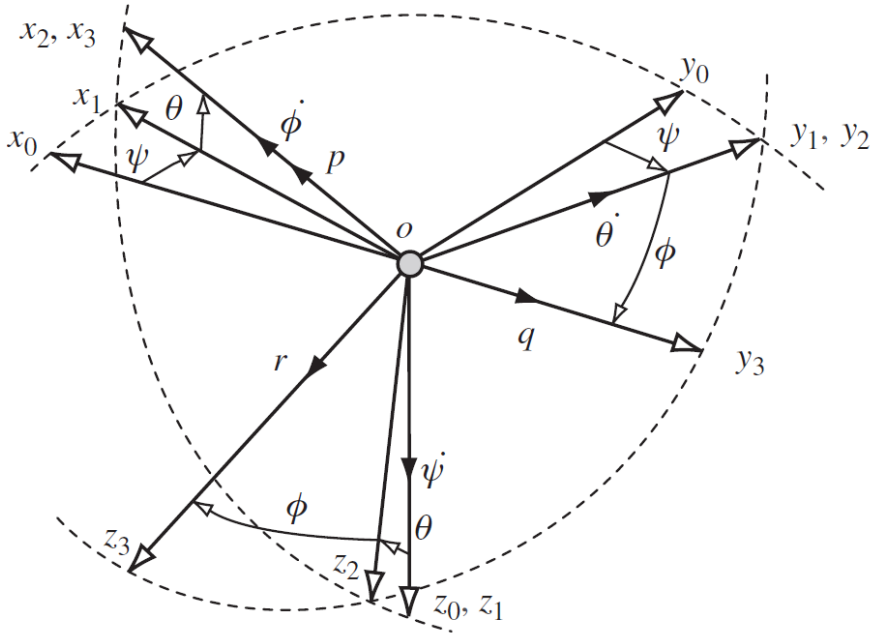
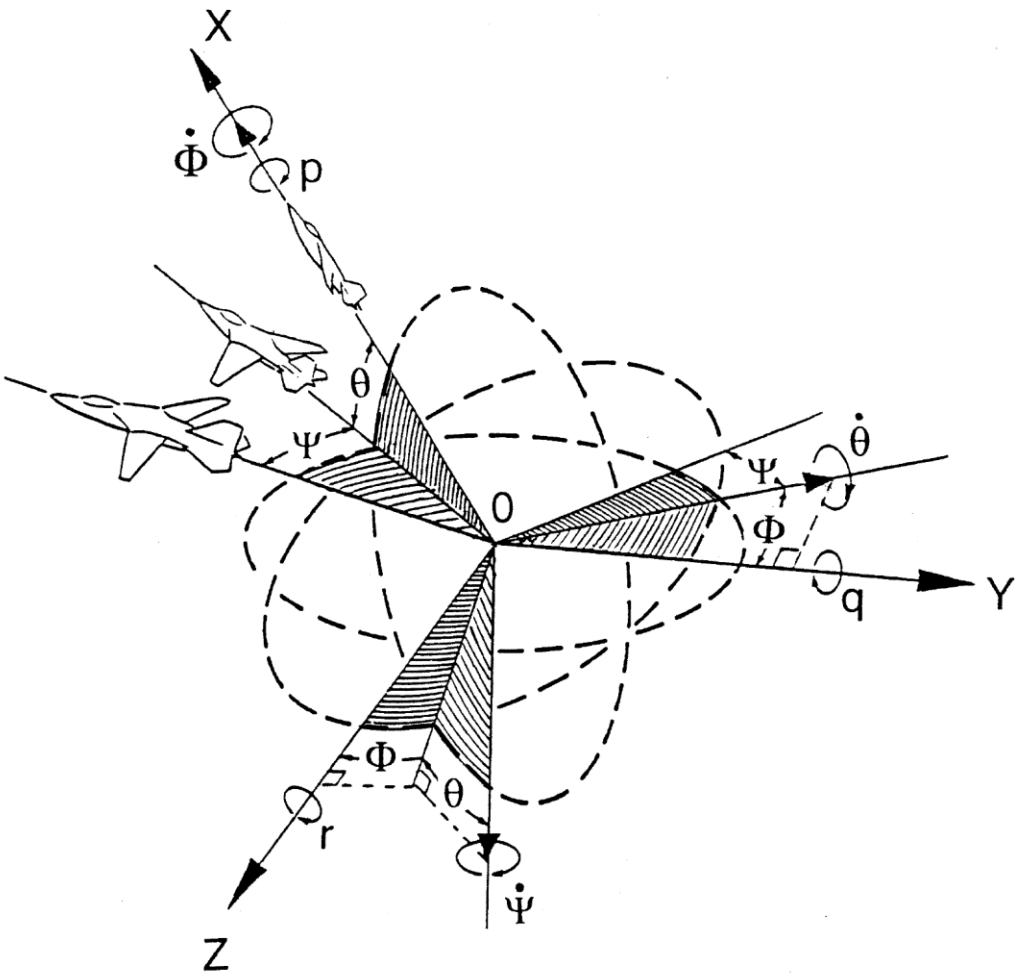


FIGURE 2.7 Angular rates transformation.

Linear Quantities Transformation

- Euler angles



Linear Quantities Transformation

- Rotation from inertial frame to body frame

$$\begin{bmatrix} ox_3 \\ oy_3 \\ oz_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} ox_0 \\ oy_0 \\ oz_0 \end{bmatrix} \tag{2.10}$$

- Direction cosine matrix **D**

$$\begin{bmatrix} ox_3 \\ oy_3 \\ oz_3 \end{bmatrix} = \mathbf{D} \begin{bmatrix} ox_0 \\ oy_0 \\ oz_0 \end{bmatrix} \qquad \begin{bmatrix} ox_0 \\ oy_0 \\ oz_0 \end{bmatrix} = \mathbf{D}^{-1} \begin{bmatrix} ox_3 \\ oy_3 \\ oz_3 \end{bmatrix}$$

$$\mathbf{D} = \begin{bmatrix} \cos\theta\cos\psi & \cos\theta\sin\psi & -\sin\theta \\ \sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi & \sin\phi\sin\theta\sin\psi + \cos\phi\cos\psi & \sin\phi\cos\theta \\ \cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi & \cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi & \cos\phi\cos\theta \end{bmatrix}$$

Angular Velocities Transformation

- The angular body rates p, q, r are shown in the aircraft axes ($ox_3y_3z_3$), then, considering each rotation in turn necessary to bring the aircraft axes into coincidence with the datum axes ($ox_0y_0z_0$).

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\sin\theta \\ 0 & \cos\phi & \sin\phi\cos\theta \\ 0 & -\sin\phi & \cos\phi\cos\theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} \tag{2.21}$$

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi\sec\theta & \cos\phi\sec\theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix} \tag{2.22}$$

Consider Euler bank angle rate, $\dot{\Phi}$

$$\text{Component of } \dot{\Phi} \text{ along} \quad \begin{cases} \text{OX} = \dot{\Phi} \\ \text{OY} = 0 \\ \text{OZ} = 0 \end{cases}$$

Consider Euler pitch angle rate, $\dot{\theta}$

$$\text{Component of } \dot{\theta} \text{ along} \quad \begin{cases} \text{OX} = 0 \\ \text{OY} = \dot{\theta} \cos \Phi \\ \text{OZ} = -\dot{\theta} \sin \Phi \end{cases}$$

Consider Euler yaw angle rate, $\dot{\psi}$

$$\text{Component of } \dot{\psi} \text{ along} \quad \begin{cases} \text{OX} = -\dot{\Psi} \sin \theta \\ \text{OY} = \dot{\Psi} \cos \theta \sin \Phi \\ \text{OZ} = \dot{\Psi} \cos \theta \cos \Phi \end{cases}$$

Hence

$$p = \dot{\Phi} - \dot{\Psi} \sin \theta \tag{3.16}$$

$$q = \dot{\theta} \cos \Phi + \dot{\Psi} \cos \theta \sin \Phi \tag{3.17}$$

$$r = \dot{\Psi} \cos \theta \cos \Phi - \dot{\theta} \sin \Phi \tag{3.18}$$

Angular Velocities Transformation

- When the aircraft perturbations are small, such that(φ, θ, ψ) may be treated as small angles, equations(2.21) and (2.22) may be approximated by

$$\begin{aligned} p &= \dot{\phi} \\ q &= \dot{\theta} \\ r &= \dot{\psi} \end{aligned} \qquad (2.23)$$

Calculation of Aircraft Trim Condition

- Defining the trim condition
  - The total axial force X is given by resolving the total lift L, total drag D, weight mg, and thrust τ_e into the ox axis; these components must sum to zero in trim.

$$X = L\sin\alpha_e + \tau_e\cos\kappa - D\cos\alpha_e - mg\sin(\alpha_e + \gamma_e) = 0 \quad (3.42)$$

$$Z = mg\cos(\alpha_e + \gamma_e) - L\cos\alpha_e - D\sin\alpha_e - \tau_e\sin\kappa = 0 \quad (3.43)$$

$$M = M_0 + L_w(h - h_0)\bar{\bar{c}} - L_T l_T + \tau_e z_\tau = 0 \quad (3.44)$$

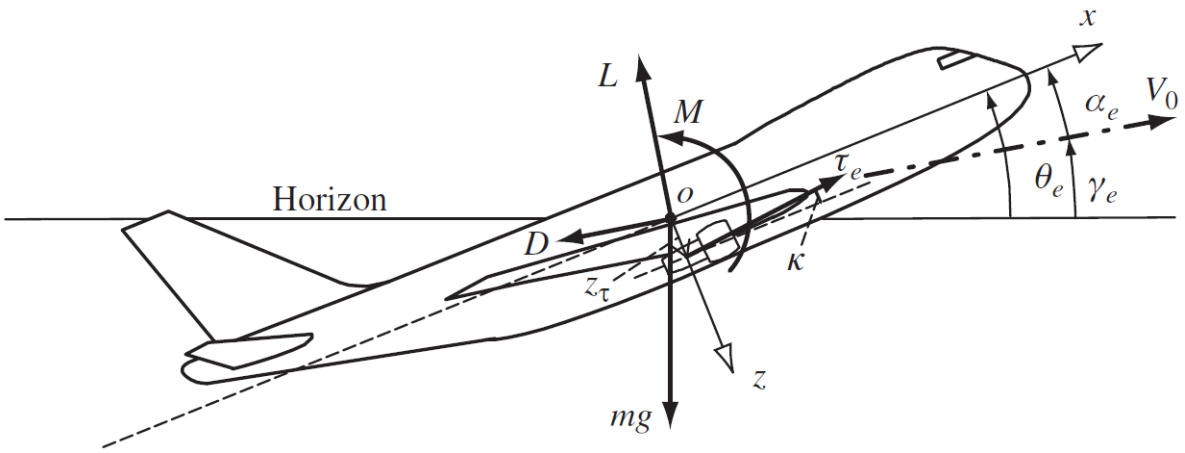


FIGURE 3.23 Symmetric forces and moments acting on a trimmed aircraft.

Steady Trim Condition

- The angular airspeeds are all zero, $P_e=Q_e=R_e=0$
- The constant airspeeds in the aircraft frame of reference are U_e and W_e, while $V_e=0$
- Now, a small perturbation is applied to obtain the equations of motion for an aircraft

$$U \rightarrow U_e + u \qquad p \rightarrow P_e + p$$
$$V \rightarrow V_e + v \qquad q \rightarrow Q_e + q$$
$$W \rightarrow W_e + w \qquad r \rightarrow R_e + r$$

  - where U_e, V_e, P_e etc describe the chosen flight condition

Equations of Motion for Small Disturbances

- Motion Variables Notation

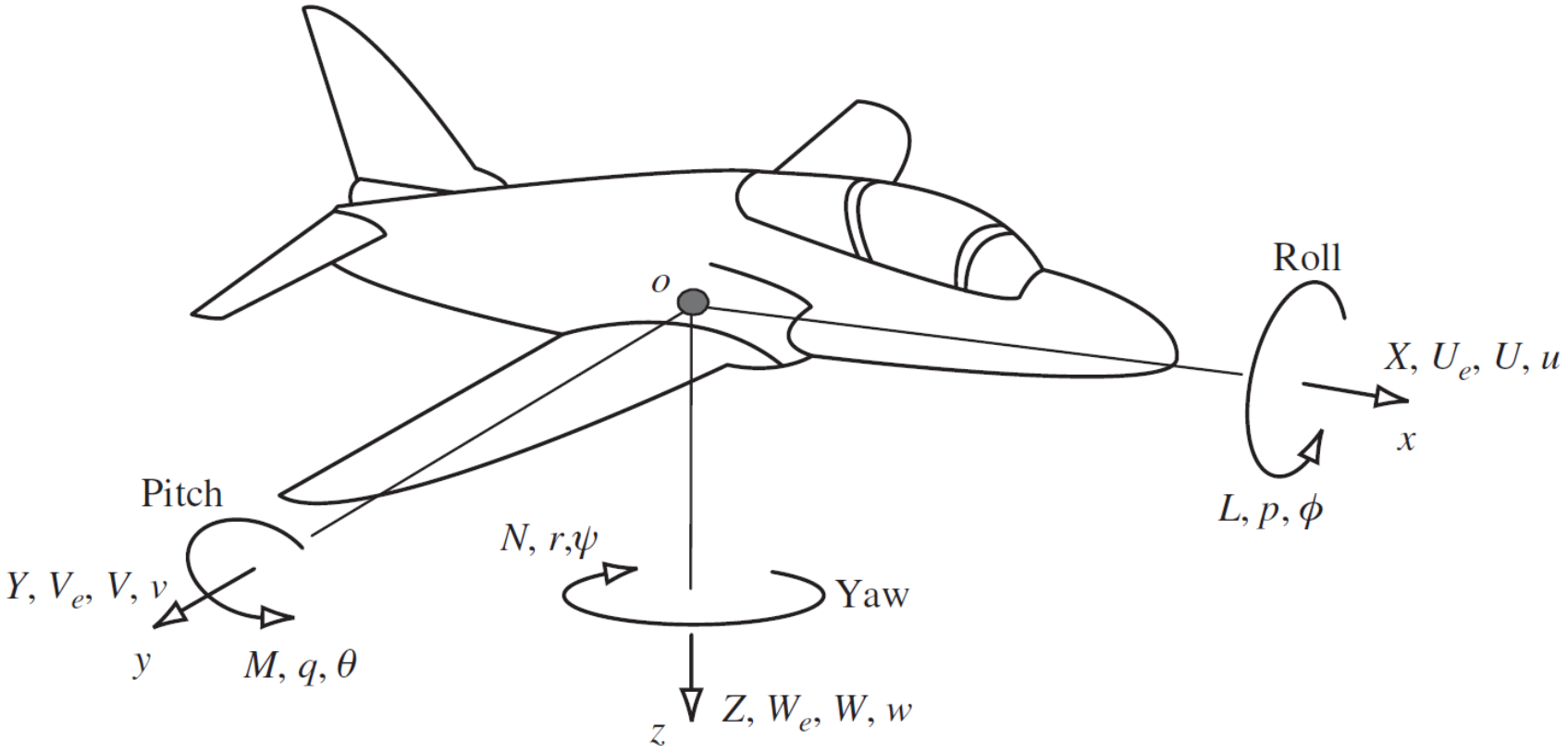


FIGURE 2.3 Motion variables notation.

Perturbation Variables

- The perturbation variables are shown in Fig. 2.3 and summarized in Table.

Table 2.1 Summary of Motion Variables

	Trimmed Equilibrium			**Perturbed**		
Aircraft axis	ox	oy	oz	ox	oy	oz
Force	0	0	0	X	Y	Z
Moment	0	0	0	L	M	N
Linear velocity	U_e	V_e	W_e	U	V	W
Angular velocity	0	0	0	p	q	r
Attitude	0	θ_e	0	ϕ	θ	ψ

Airplane Geometry

- A simple description of the perturbation variables is given in Table 2.2.

Table 2.2 Perturbation Variables

X	Axial "drag" force	Sum of the components of aerodynamic, thrust and weight forces
Y	Side force	
Z	Normal "lift" force	
L	Rolling moment	Sum of the components of aerodynamic, thrust and weight moments
M	Pitching moment	
N	Yawing moment	
p	Roll rate	Components of angular velocity
q	Pitch rate	
r	Yaw rate	
U	Axial velocity	Total linear velocity components of the cg
V	Lateral velocity	
W	Normal velocity	

Equations of motion for a rigid symmetric aircraft

- The derivation of the equations of motion is based on the application of Newton's second law of motion, which states,
  - *The acceleration of a particle acted on by a force is proportional to and in the direction of the force, the factor of proportionality being independent of the force and the time.*
- Two sets of equations are derived:
  - Sum of forces acting on the system (internal and external) are equal to its mass times its acceleration
  - Sum of moments acting on the system (internal and external) are equal to its moment of inertia times its angular acceleration

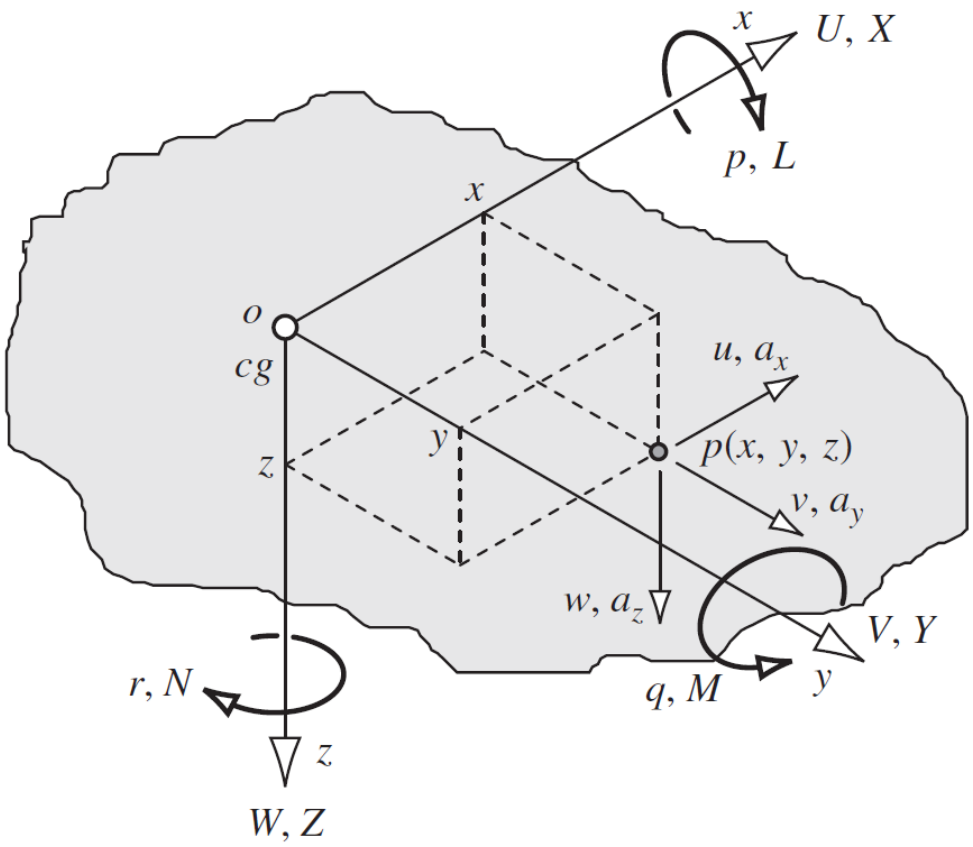
$$\begin{aligned} \text{Force} &= \text{Mass} \times \text{Inertial Acceleration} \\ \text{Moment} &= \text{Moment of Inertia} \times \text{Angular Acceleration} \end{aligned} \tag{4.1}$$

Equations of motion for a rigid symmetric aircraft

- Developing the equations of motion
  - Therefore, the object of the derivation is to estimate the accelerations (linear and angular of the aircraft)
  - As usual, the same equations of motion can be obtained using Lagrange's equation (i.e. conservation of energy)

Equations of motion for a rigid symmetric aircraft

- Body and axes
- Axis system
  - Could be any body but in this case it is an aircraft of mass m.
  - For the moment it is a flexible body Any point p on the body can have a velocity and acceleration with respect to the c.g.



Motion referred to generalised body axes.

Generalized Force Equations

- Consider now an incremental mass δm at point $p(x,y,z)$ in the rigid body.

$$\Sigma \delta m a'_x = X$$
$$\Sigma \delta m a'_y = Y$$
$$\Sigma \delta m a'_z = Z$$

  - Note that since the origin of axes coincides with the c.g.,

$$\Sigma \delta m x = \Sigma \delta m y = \Sigma \delta m z = 0$$

- The resultant components of total force acting on the rigid body are given by

$$m(\dot{U} - rV + qW) = X$$
$$m(\dot{V} - pW + rU) = Y$$
$$m(\dot{W} - qU + pV) = Z$$

  - where m is the total mass of the body

Generalized Moment Equations

- Moment Equations could be rewritten

$$I_x\dot{p} - (I_y - I_z)qr + I_{xy}(pr - \dot{q}) - I_{xz}(pq + \dot{r}) + I_{yz}(r^2 - q^2) = L$$

$$I_y\dot{q} + (I_x - I_z)pr + I_{yz}(pq - \dot{r}) + I_{xz}(p^2 - r^2) - I_{xy}(qr + \dot{p}) = M$$

$$I_z\dot{r} - (I_x - I_y)pq - I_{yz}(pr + \dot{q}) + I_{xz}(qr - \dot{p}) + I_{xy}(q^2 - p^2) = N$$

Moments and Products of Inertia

$I_x = \Sigma\delta m(y^2 + z^2)$	Moment of inertia about ox axis
$I_y = \Sigma\delta m(x^2 + z^2)$	Moment of inertia about oy axis
$I_z = \Sigma\delta m(x^2 + y^2)$	Moment of inertia about oz axis
$I_{xy} = \Sigma\delta mxy$	Product of inertia about ox and oy axes
$I_{xz} = \Sigma\delta mxz$	Product of inertia about ox and oz axes
$I_{yz} = \Sigma\delta myz$	Product of inertia about oy and oz axes

Generalized Moment Equations

- When the generalized body represents an aircraft, the moment equations may be simplified since it is assumed that the aircraft is symmetric about the *oxz* plane and that the mass is uniformly distributed.
- As a result, the products of inertia $I_{xy} = I_{yz} = 0$.
- Thus the moment equations simplify to the following:

$$
\begin{aligned}
I_x\dot{p} - (I_y - I_z)qr - I_{xz}(pq + \dot{r}) &= L \\
I_y\dot{q} + (I_x - I_z)pr + I_{xz}(p^2 - r^2) &= M \\
I_z\dot{r} - (I_x - I_y)pq + I_{xz}(qr - \dot{p}) &= N
\end{aligned}
$$

Discussion of the Equations

- If we can solve for U, V, W, p, q, r as functions of time, then we know the complete time history of the motion of the aircraft.
- Unfortunately, terms such as rU, pV, qW, etc and pq, r^2, qr etc are nonlinear.
- Furthermore, we have only defined the inertial loads up to now.
- We have not said anything about the external loads acting on the aircraft.

Nonlinear vs. Linear

- **Nonlinear equations of motion**
  - Compute "exact" flight paths and motions
    - Simulate flight motions
    - Optimize flight paths
    - Predict performance
  - Provide basis for approximate solutions

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{f}\left[\mathbf{x}(t),\mathbf{u}(t),\mathbf{w}(t),\mathbf{p}(t),t\right]$$

- **Linear equations of motion**
  - Simplify computation of flight paths and solutions
  - Define modes of motion
  - Provide basis for control system design and flying qualities analysis

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{F}\,\mathbf{x}(t) + \mathbf{G}\,\mathbf{u}(t) + \mathbf{L}\,\mathbf{w}(t)$$

Disturbance Forces and Moments

- The traditional approach, after Bryan (1911), is to assume that the disturbing forces and moments are due to *a*erodynamic effects, *g*ravitational effects, movement of aerodynamic *c*ontrols, *p*ower effects and the effects of atmospheric *d*isturbances.
- There are five sources of external forces and moments:
  - **A**erodynamic
  - **G**ravitational
  - **C**ontrols
  - **P**ropulsion
  - Atmospheric **D**isturbances

Disturbance Forces and Moments

- The full equations of motion in the presence of external forces and moments are

$$
\begin{aligned}
m(\dot{U} - rV + qW) &= X_a + X_g + X_c + X_p + X_d \\
m(\dot{V} - pW + rU) &= Y_a + Y_g + Y_c + Y_p + Y_d \\
m(\dot{W} - qU + pV) &= Z_a + Z_g + Z_c + Z_p + Z_d \\
I_x\dot{p} - (I_y - I_z)qr - I_{xz}(pq + \dot{r}) &= L_a + L_g + L_c + L_p + L_d \\
I_y\dot{q} + (I_x - I_z)pr + I_{xz}(p^2 - r^2) &= M_a + M_g + M_c + M_p + M_d \\
I_z\dot{r} - (I_x - I_y)pq + I_{xz}(qr - \dot{p}) &= N_a + N_g + N_c + N_p + N_d
\end{aligned}
\tag{4.20}
$$

- The equations are *non-linear* and their solution by analytical means is not generally practicable.
- We will investigate each one of these separately

Aerodynamic Terms

- An explicit calculation of the aerodynamic interactions due to a disturbance is very complex.
- An alternative is to focus on small perturbations around a known set of aerodynamic forces, X_{ae}, Y_{ae}, Z_{ae}, L_{ae}, M_{ae}, N_{ae}.
- This approach was first used in 1911 and is still in use today. The method first described by Bryan.
- Although the approach is unchanged, the rather more modern notation of Hopkin (1970) is adopted.

Aerodynamic Perturbations

- Denote $\mathbf{F}=[X\ Y\ Z\ L\ M\ N]^{\mathrm{T}}$.
- Denote $\mathbf{v}=[u\ v\ w\ p\ q\ r]^{\mathrm{T}}$.
- Express the aerodynamic forces as a Taylor series around the trim aerodynamic force $\mathbf{F}_{ae}$.

$$\mathbf{F}_a = \mathbf{F}_{a_e} + \frac{\partial \mathbf{F}}{\partial \mathbf{v}}\mathbf{v} + \frac{1}{2!}\frac{\partial^2 \mathbf{F}}{\partial \mathbf{v}^2}\mathbf{v}^2 + \frac{1}{3!}\frac{\partial^3 \mathbf{F}}{\partial \mathbf{v}^3}\mathbf{v}^3 + \cdots$$
$$+ \frac{\partial \mathbf{F}}{\partial \dot{\mathbf{v}}}\dot{\mathbf{v}} + \frac{1}{2!}\frac{\partial^2 \mathbf{F}}{\partial \dot{\mathbf{v}}^2}\dot{\mathbf{v}}^2 + \frac{1}{3!}\frac{\partial^3 \mathbf{F}}{\partial \dot{\mathbf{v}}^3}\dot{\mathbf{v}}^3 + \cdots$$

Jacobians, Hessians

- The first order derivative (Jacobian) is defined as

$$\frac{\partial \mathbf{F}}{\partial \mathbf{v}} = \begin{bmatrix} \frac{\partial X}{\partial u} & \frac{\partial X}{\partial v} & \cdots & \frac{\partial X}{\partial r} \\ \frac{\partial Y}{\partial u} & \frac{\partial Y}{\partial v} & \cdots & \frac{\partial Y}{\partial r} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial N}{\partial u} & \frac{\partial N}{\partial v} & & \frac{\partial N}{\partial r} \end{bmatrix}$$

- The second order derivative (Hessian) and higher orders are defined similarly
- The Jacobian and higher orders with respect to are also defined similarly.

$$X_a = X_{a_e} + \left(\frac{\partial X}{\partial u} u + \frac{\partial^2 X}{\partial u^2} \frac{u^2}{2!} + \frac{\partial^3 X}{\partial u^3} \frac{u^3}{3!} + \frac{\partial^4 X}{\partial u^4} \frac{u^4}{4!} + \ldots\ldots\ldots\ldots\ldots \right)$$

$$+ \left(\frac{\partial X}{\partial v} v + \frac{\partial^2 X}{\partial v^2} \frac{v^2}{2!} + \frac{\partial^3 X}{\partial v^3} \frac{v^3}{3!} + \frac{\partial^4 X}{\partial v^4} \frac{v^4}{4!} + \ldots\ldots\ldots\ldots\ldots \right)$$

$$+ \left(\frac{\partial X}{\partial w} w + \frac{\partial^2 X}{\partial w^2} \frac{w^2}{2!} + \frac{\partial^3 X}{\partial w^3} \frac{w^3}{3!} + \frac{\partial^4 X}{\partial w^4} \frac{w^4}{4!} + \ldots\ldots\ldots\ldots\ldots \right)$$

$$+ \left(\frac{\partial X}{\partial p} p + \frac{\partial^2 X}{\partial p^2} \frac{p^2}{2!} + \frac{\partial^3 X}{\partial p^3} \frac{p^3}{3!} + \frac{\partial^4 X}{\partial p^4} \frac{p^4}{4!} + \ldots\ldots\ldots\ldots\ldots \right)$$

$$+ \left(\frac{\partial X}{\partial q} q + \frac{\partial^2 X}{\partial q^2} \frac{q^2}{2!} + \frac{\partial^3 X}{\partial q^3} \frac{q^3}{3!} + \frac{\partial^4 X}{\partial q^4} \frac{q^4}{4!} + \ldots\ldots\ldots\ldots\ldots \right)$$

$$+ \left(\frac{\partial X}{\partial r} r + \frac{\partial^2 X}{\partial r^2} \frac{r^2}{2!} + \frac{\partial^3 X}{\partial r^3} \frac{r^3}{3!} + \frac{\partial^4 X}{\partial r^4} \frac{r^4}{4!} + \ldots\ldots\ldots\ldots\ldots \right)$$

$$+ \left(\frac{\partial X}{\partial \dot{u}} \dot{u} + \frac{\partial^2 X}{\partial \dot{u}^2} \frac{\dot{u}^2}{2!} + \frac{\partial^3 X}{\partial \dot{u}^3} \frac{\dot{u}^3}{3!} + \ldots\ldots\ldots\ldots\ldots \right)$$

$$+ \left(\frac{\partial X}{\partial \dot{v}} \dot{v} + \frac{\partial^2 X}{\partial \dot{v}^2} \frac{\dot{v}^2}{2!} + \frac{\partial^3 X}{\partial \dot{v}^3} \frac{\dot{v}^3}{3!} + \ldots\ldots\ldots\ldots\ldots \right)$$

$$+ \text{series terms in } \dot{w}, \dot{p}, \dot{q}, \text{ and } \dot{r}$$

$$+ \text{series terms in higher order derivatives}$$

Simplifications

- Only first order terms are retained.
- Derivatives with respect to and are rarely encountered. Only derivatives with respect to are retained.
- The aerodynamic forces and moments then are given by

$$\mathbf{F}_a = \mathbf{F}_{a_e} + \frac{\partial \mathbf{F}}{\partial \mathbf{v}}\mathbf{v} + \frac{\partial \mathbf{F}}{\partial \dot{w}}\dot{w}$$

Aerodynamic Stability Derivatives

- The terms $\partial X/\partial u$, $\partial L/\partial v$ etc are known as aerodynamic stability derivatives
- The aerodynamic forces and moments are usually written as, e.g.

$$X_a = X_{a_e} + \frac{\partial X}{\partial u}u + \frac{\partial X}{\partial v}v + \frac{\partial X}{\partial w}w + \frac{\partial X}{\partial p}p + \frac{\partial X}{\partial q}q + \frac{\partial X}{\partial r}r + \frac{\partial X}{\partial \dot{w}}\dot{w} \qquad (4.29)$$

$$X_a = X_{a_e} + \overset{\circ}{X}_u u + \overset{\circ}{X}_v v + \overset{\circ}{X}_w w + \overset{\circ}{X}_p p + \overset{\circ}{X}_q q + \overset{\circ}{X}_r r + \overset{\circ}{X}_{\dot{w}} \dot{w} \qquad (4.30)$$

- Etc for the other forces and moments
- In this sense, the equations of motion of the aircraft can be written if the aerodynamic derivatives are known.
- The dressing ($\circ$) denotes the derivatives to be *dimensional*.

Gravitational Terms

- Since the origin of the airplane body axes is coincident with the cg, there is no weight moment about any axis. Therefore,

$$L_g = M_g = N_g = 0$$

- As the airplane is flying with wings level:

$$\begin{bmatrix} X_{g_e} \\ Y_{g_e} \\ Z_{g_e} \end{bmatrix} = \begin{bmatrix} -mg \sin\theta_e \\ 0 \\ mg \cos\theta_e \end{bmatrix}$$

- The angular velocity perturbations p, q, r will cause angular displacements φ, θ, ψ.
- Then, the forces will become a rotation of the
- force vector $[X_{ge}\ Y_{ge}\ Z_{ge}]^T$.

Gravitational Terms

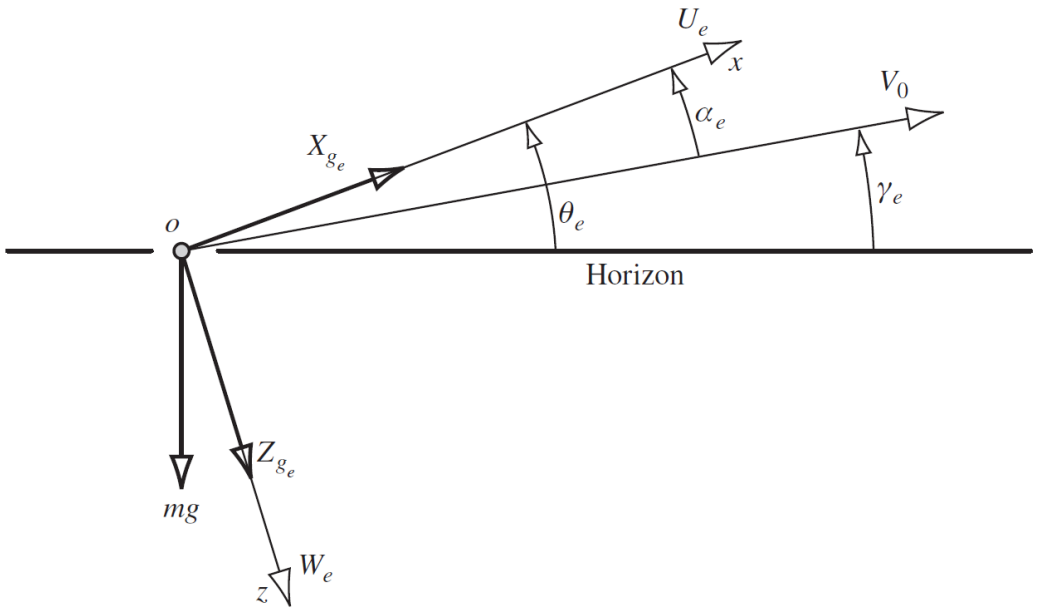


FIGURE 4.4 Steady state weight components in the plane of symmetry.

Direction Cosine Matrix

- Notice that the combined rotation in the previous slide is only correct if the aircraft first rotates in the φ direction, then in the θ and then in the ψ.
- If the order or rotations is changed, then the combined rotation matrix will also change, i.e. $R_{\varphi\theta\psi} \neq R_{\psi\varphi\theta}$.
- However, as the angles φ, θ and ψ are perturbations, they are small.
- Then, $\sin\varphi=\varphi$, $\cos\theta=1$ etc and products of perturbations can be neglected.
- Now, the combined rotation matrix becomes insensitive to rotation order!

Direction Cosine Matrix

- These rotations mean that the gravity forces are:

$$\begin{bmatrix} X_g \\ Y_g \\ Z_g \end{bmatrix} = \begin{bmatrix} 1 & \psi & -\theta \\ -\psi & 1 & \phi \\ \theta & -\phi & 1 \end{bmatrix} \begin{bmatrix} X_{g_e} \\ Y_{g_e} \\ Z_{g_e} \end{bmatrix} = \begin{bmatrix} 1 & \psi & -\theta \\ -\psi & 1 & \phi \\ \theta & -\phi & 1 \end{bmatrix} \begin{bmatrix} -mg \sin\theta_e \\ 0 \\ mg \cos\theta_e \end{bmatrix} \tag{4.26}$$

- Note that θ_e is not necessarily small, so that:

$$\begin{aligned} X_g &= -mg \sin\theta_e - mg\,\theta \cos\theta_e \\ Y_g &= mg\,\psi \sin\theta_e + mg\,\phi \cos\theta_e \\ Z_g &= mg \cos\theta_e - mg\,\theta \sin\theta_e \end{aligned} \tag{4.27}$$

Aerodynamic Control Terms

- The primary aerodynamic control surfaces are the ailerons, elevator and rudder.
- Define their deflections are ξ, η, ζ, with respect to trim positions ξ_e, η_e, ζ_e.
- The control surfaces can only apply moments, not forces.
- Define $\mathbf{M}=[L\ M\ N]^T$.
- Define $\mathbf{s}=[\xi\ \eta\ \zeta]^T$.

Aerodynamic Control Terms

- Control deflections will cause a control moment $\mathbf{M}_c$ with respect to the trim position of the form

$$M_c = \frac{\partial M}{\partial \xi}\xi + \frac{\partial M}{\partial \eta}\eta + \frac{\partial M}{\partial \zeta}\zeta$$

- Again, a more usual notation is, e.g.

$$M_c = \overset{\circ}{M}_{\xi}\,\xi + \overset{\circ}{M}_{\eta}\,\eta + \overset{\circ}{M}_{\zeta}\,\zeta$$

i.e., in terms of the aerodynamic control derivatives.

Power Terms

- For turbojet engines power is described by the thrust provided by the engine.
- Thrust has an effect on all forces and moments around the aircraft.
- The forces and moments caused by a change of thrust, τ, with respect to a trim setting, τ_e, is given by

$$Z_p = \overset{\circ}{Z}_\tau \tau$$

Linearized Equations of Motion

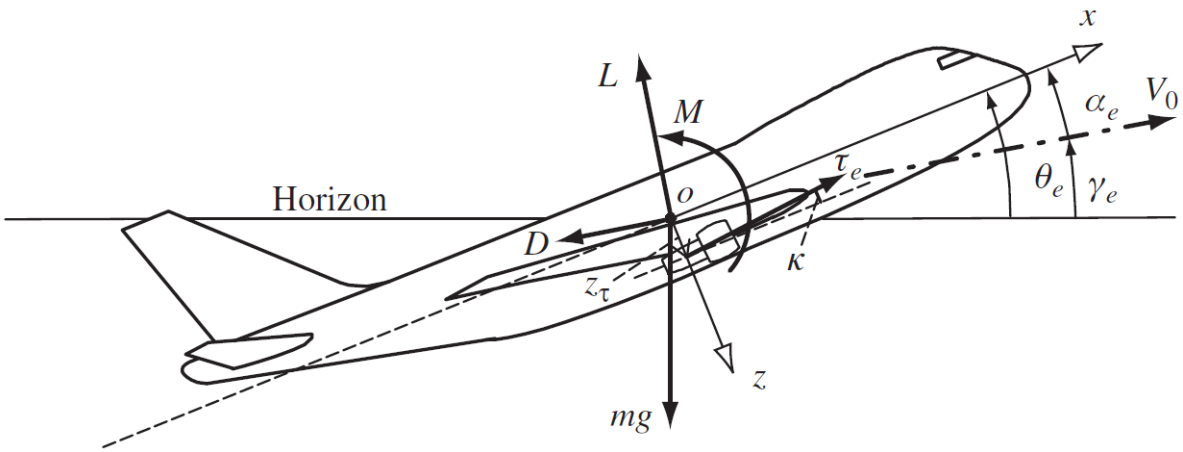
- The full equations of motion derived in the previous page are clearly nonlinear and, therefore, difficult to solve.
- The equations would be much more practical if they were linear.
- So why not linearize them?
- The linearization is performed around a chosen flight condition
- The degrees of freedom are written as

$$U \rightarrow U_e + u \qquad p \rightarrow P_e + p$$
$$V \rightarrow V_e + v \qquad q \rightarrow Q_e + q$$
$$W \rightarrow W_e + w \qquad r \rightarrow R_e + r$$

- Where U_e, V_e, P_e etc describe the chosen flight condition

Perturbed Flight Condition

- Assume rectilinear flight
- The angular airspeeds are all zero, $P_e=Q_e=R_e=0$



- The constant airspeeds in the aircraft frame of reference are U_e and W_e, while $V_e=0$
- Now, a small perturbation is applied

Linearized Equations of Motion

- A small perturbation can be described in the unperturbed frame of reference as:

$$U \rightarrow U_e + u$$

$$V \rightarrow V_e + v$$

$$W \rightarrow W_e + w$$

- As well as angular perturbations p, q, r.
- Notice that the angular velocities are all perturbations; the steady state has no angular velocity.

Linearized Equations of Motion

- Substitute these expressions to the equations of motion to obtain:

$$
\begin{aligned}
m(\dot{u} + qW_e) &= X_a + X_g + X_c + X_p \\
m(\dot{v} - pW_e + rU_e) &= Y_a + Y_g + Y_c + Y_p \\
m(\dot{w} - qU_e) &= Z_a + Z_g + Z_c + Z_p \\
I_x\dot{p} - I_{xz}\dot{r} &= L_a + L_g + L_c + L_p \\
I_y\dot{q} &= M_a + M_g + M_c + M_p \\
I_z\dot{r} - I_{xz}\dot{p} &= N_a + N_g + N_c + N_p
\end{aligned}
$$

- Where terms in, say *uv* or *qr* or *wq* are neglected and, of course, U_e, W_e are constants
- Atmospheric disturbances are also assumed negligible

$$X_d = Y_d = Z_d = L_d = M_d = N_d = 0$$

State-Space Form

- These very long and complex equations can be written in the standard compact state-space form as:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

  - **x(*t*)** is the column vector of n state variables, called the *state vector*
  - **u(*t*)** is the column vector of m input variables, called the *input vector*
  - **A** is the(n x n) state matrix
  - **B** is the(n x m) input matrix

where

$\mathbf{x}=[u\ v\ w\ p\ q\ r\ \varphi\ \theta\ \psi]^T$ and

$\mathbf{u}=[\xi\ \eta\ \zeta\ \tau]^T$

State-Space Form

- For aircraft the output variables (measurements) are usually chosen to be the state variables so the output equation in the state space form is simply

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t)$$

where

- **y(t)** is the column vector of r output variables, called the *output vector*
- **C** is the(r x n) output matrix
- **D** is the(r x m) direct matrix

- For most airplane problems it is convenient to choose the output variables to be the state variables. Thus

$$\mathbf{y}(t) = \boldsymbol{x}(t) \quad \text{and} \quad r = n$$

and consequently

C = I, the(n x n) identity matrix

D = 0, the(n x m) zero matrix

Discussion

- The equations of motion of a complete aircraft then, about a trim position, are 9 first order linear differential equations.
- They do not include the effects of, say, trim tabs or flaps but these can be added in the usual way.
- The aerodynamic stability derivatives and the control derivatives must be known, otherwise the equations of motion are useless.
- These derivatives are usually measured in flight tests.

Decoupling

- When studying static stability we stated that usually pitch stability is decoupled from roll and yaw stability.
- The same holds true, for many aircraft, in the dynamic case.
- Therefore, the equations of motion are usual split into two sets:
  - Longitudinal equations of motion (pitch)
  - Lateral equations of motion (yaw and roll)

Longitudinal EoM

- The motion is described only by the axial force X, normal force, Z and pitching moment M.
- The lateral motion variables v, p and r are zero, as are the rotations φ and ψ.
- Aileron and rudder deflection do not usually affect pitching.

Longitudinal Forces and Moments

- The main forces and moments acting along and about the OX, OY, OZ axes affecting longitudinal motion are set out below:

$$X_a = X_u u + X_w w \tag{3.28}$$

$$Z_a = Z_u u + Z_w w + Z_{\dot{w}} \dot{w} + Z_q q + Z_\eta \eta \tag{3.29}$$

$$M = M_w w + M_q q + M_{\dot{w}} \dot{w} + M_\eta \eta \tag{3.30}$$

Lateral Forces and Moments

- The main sideways force, *Y*a, the rolling moment, *L*, and the yawing moment, *N*, resulting from changes in the sideslip velocity, *v*, rate of roll, *p*, and rate of yaw, *r*, from the steady flight condition are set out below. The equations of motion become:

$$Y_a = Y_v v + Y_\zeta \zeta \tag{3.31}$$

$$L = L_v v + L_p p + L_r r + L_\xi \xi + L_\zeta \zeta \tag{3.32}$$

$$N = N_v v + N_p p + N_r r + N_\xi \xi + N_\zeta \zeta \tag{3.33}$$

Equations of Longitudinal and Lateral Motion

- The equations of longitudinal motion for small disturbances then become:

$$X_u u + X_w w - mg\theta = m\,(\dot{u} + W_0 q) \tag{3.34}$$

$$Z_u u + Z_w w + Z_{\dot{w}}\dot{w} + Z_q q + Z_\eta \eta = m(\dot{w} - U_0 q)) \tag{3.35}$$

$$M_w w + M_q q + M_{\dot{w}}\dot{w} + M_\eta \eta = I_y \dot{q} \tag{3.36}$$

- The equations of lateral motion for small disturbances then become:

$$Y_v v + Y_\zeta \zeta + mg\Phi = m(\dot{v} - W_0 p + U_0 r) \tag{3.37}$$

$$L_v v + L_p p + L_r r + L_\zeta \zeta + L_\xi \xi = I_x \dot{p} \tag{3.38}$$

$$N_v v + N_p p + N_r r + N_\zeta \zeta + N_\xi \xi = I_z \dot{r} \tag{3.39}$$

State-Space Form

$$\begin{bmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \frac{X_u}{m} & \frac{X_w}{m} & -W_0 & -g \\ \frac{Z_u}{m} & \frac{Z_w}{m} & U_0 & 0 \\ 0 & \frac{M_w}{I_y} & \frac{M_q}{I_y} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{z_\eta}{m} \\ \frac{M_\eta}{I_y} \\ 0 \end{bmatrix} [\eta] \qquad (3.42)$$

$$\begin{bmatrix} \dot{v} \\ \dot{p} \\ \dot{r} \\ \dot{\Phi} \end{bmatrix} = \begin{bmatrix} \frac{Y_v}{m} & W_0 & -U_0 & g \\ \frac{L_v}{I_x} & \frac{L_p}{I_x} & \frac{L_r}{I_x} & 0 \\ \frac{N_v}{I_z} & \frac{N_p}{I_z} & \frac{N_r}{I_z} & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} v \\ p \\ r \\ \Phi \end{bmatrix} + \begin{bmatrix} \frac{Y_\zeta}{m} & 0 \\ \frac{L_\zeta}{I_x} & \frac{L_\xi}{I_x} \\ \frac{N_\zeta}{I_z} & \frac{N_\xi}{I_z} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \zeta \\ \xi \end{bmatrix} \qquad (3.43)$$

State-Space Form

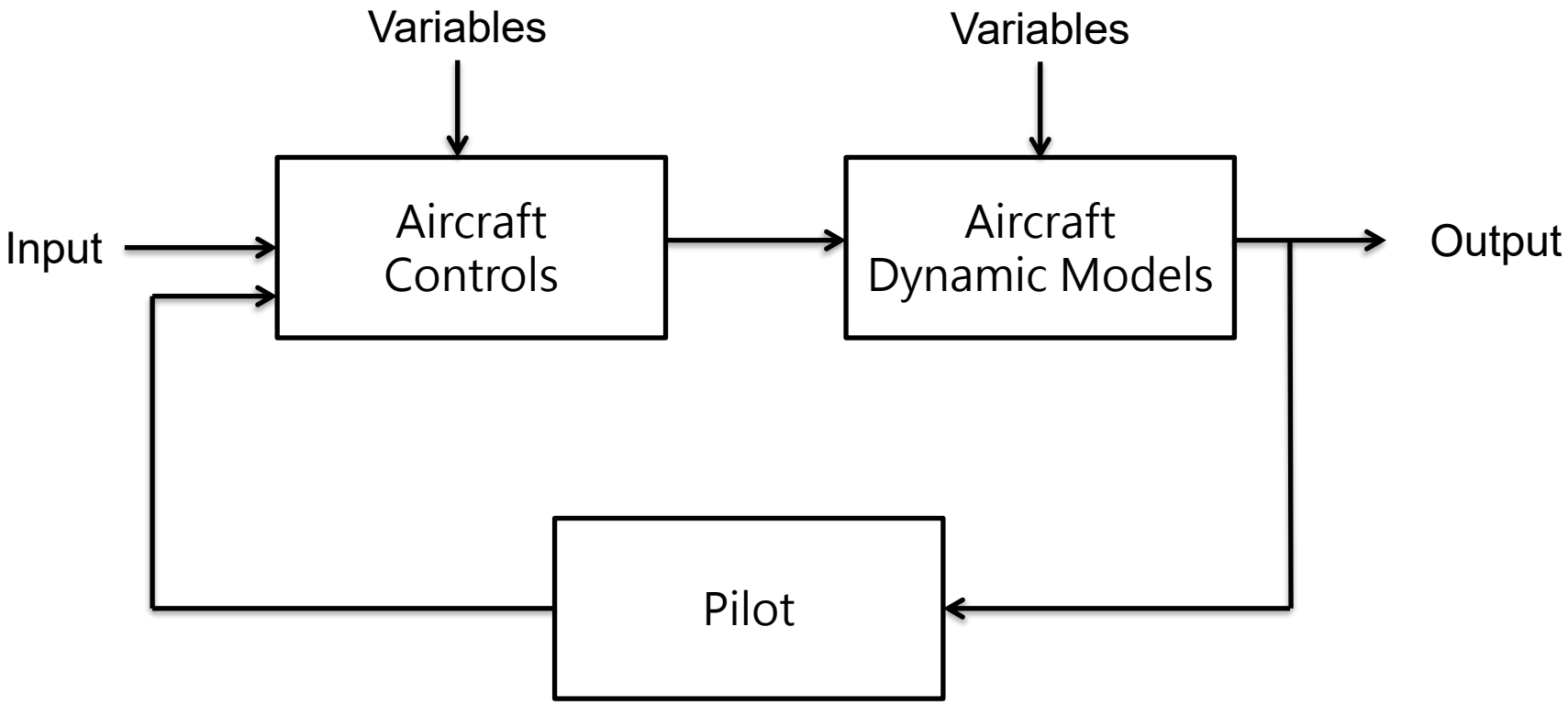
- A sea level flight condition of a Lockheed F-104 Starfighter and its longitudinal equation of motion is as follows:

$$\begin{bmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} -0.0352 & 0.1070 & 0 & -32.2 \\ -0.2140 & -0.4400 & 305 & 0 \\ 1.198 \times 10^{-4} & -0.0154 & -0.4498 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix} + \begin{bmatrix} 0 \\ -22.1206 \\ -4.6580 \\ 0 \end{bmatrix} \eta$$

- Transfer functions of α(s)/η(s)

$$\frac{\alpha(s)}{\eta(s)} = \frac{w\,(s)/U_e}{\eta(s)} = \frac{1}{305}\,\frac{-22.12s - 1431}{s^2 + 0.8898s + 4.895}$$

Closed Loop Flight Control

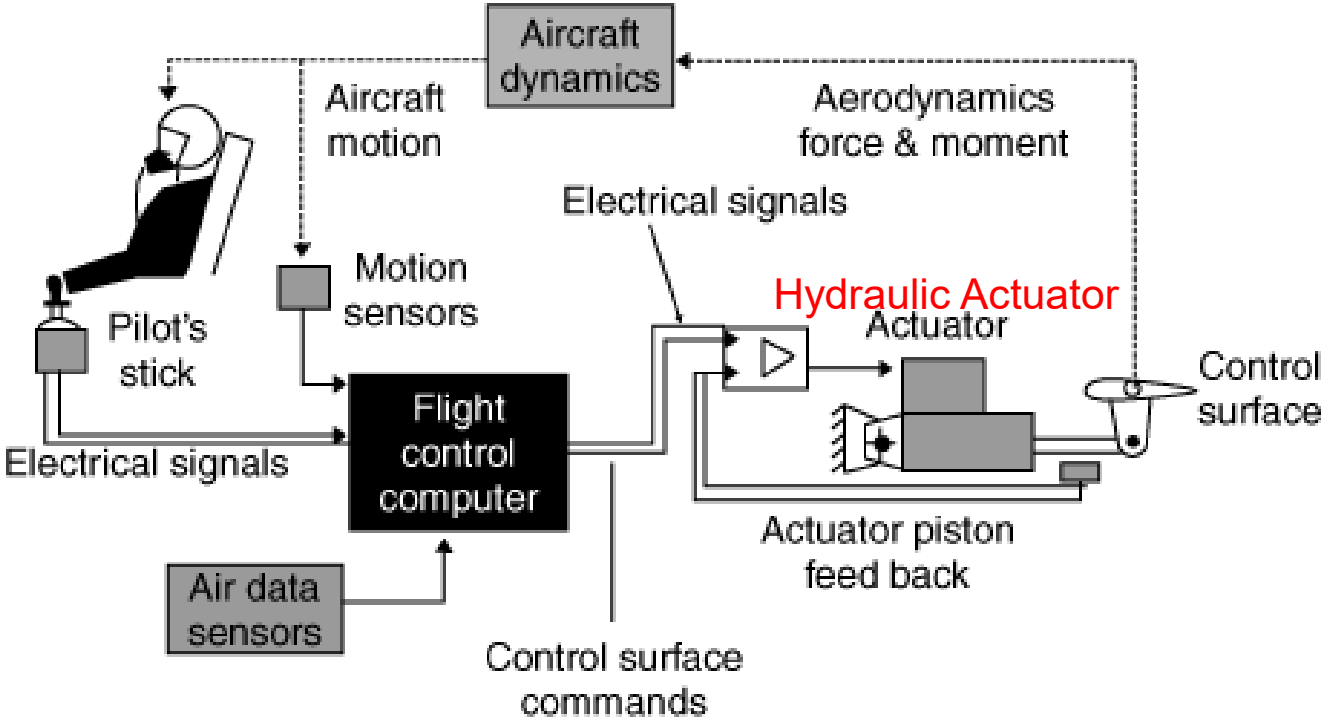


Pilot's Position

- Pilot does actually play as a real time feedback system, including sensing of touch, vision, acoustics and directional sense, with precision computer of thinking and experience, from fast actuators of hands and legs for manipulation
- But, a pilot may not be able to respond fast changes of flight condition. Pilots need computer control assistances

Pilot and Flight Control System

- Pilot commands all terminal elements via electronic signals, no actual force is required



Longitudinal Control and Response

- To manoeuvre in the longitudinal (or pitch) plane, the pilot controls the tailplane/elevator angle and hence the pitching moment exerted about the CG by the tailplane lift.

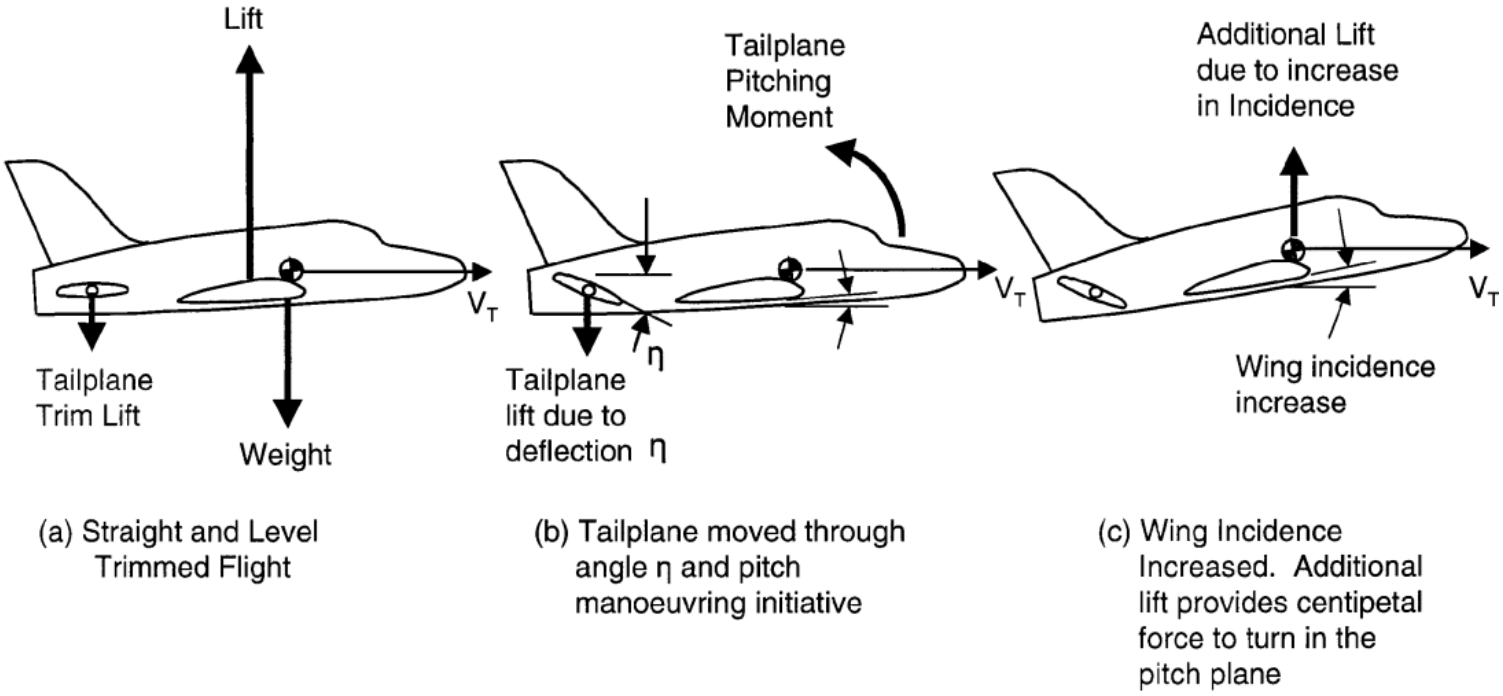


Fig. 3.24 Manoeuvring in the pitch plane.

Pitch Rate Response to Elevator Angle

- The aircraft's pitch rate, q, to the tailplane/elevator angular movement, η, from the trimmed position.

$$\frac{q}{\eta} = \frac{K(D^3 + b_2 D^2 + b_1 D + b_0)}{(D^4 + a_3 D^3 + a_2 D^2 + a_1 D + a_0)}$$

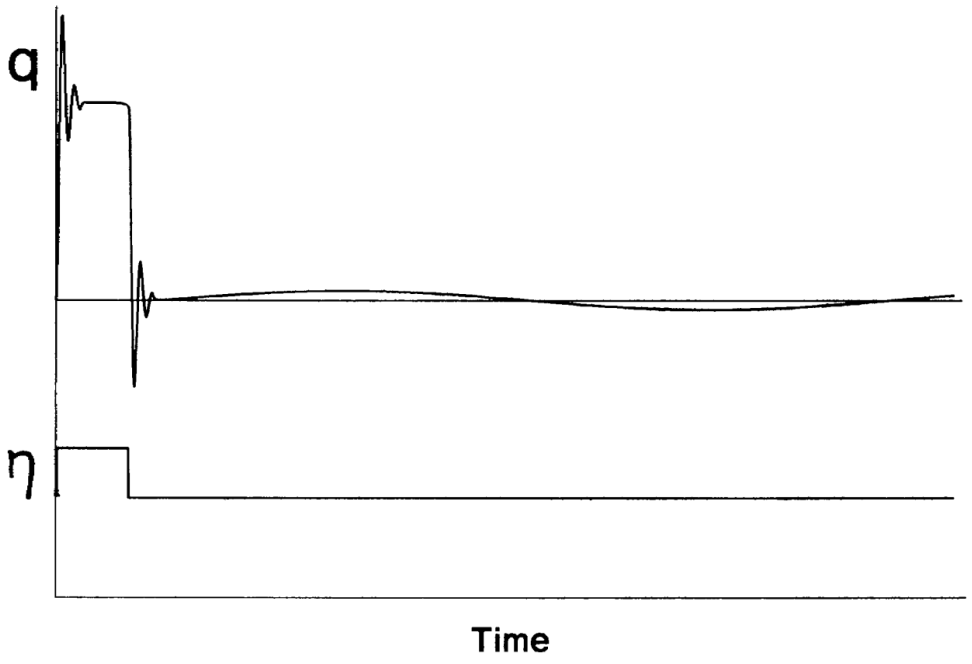


Fig. 3.25 Pitch response.

AoA Response to Elevator Angle

$$\alpha = \frac{w}{U} \tag{3.46}$$

$$\frac{\alpha}{\eta} = \frac{K T_2}{(D^2 + A_1 D + A_2)} \tag{3.60}$$

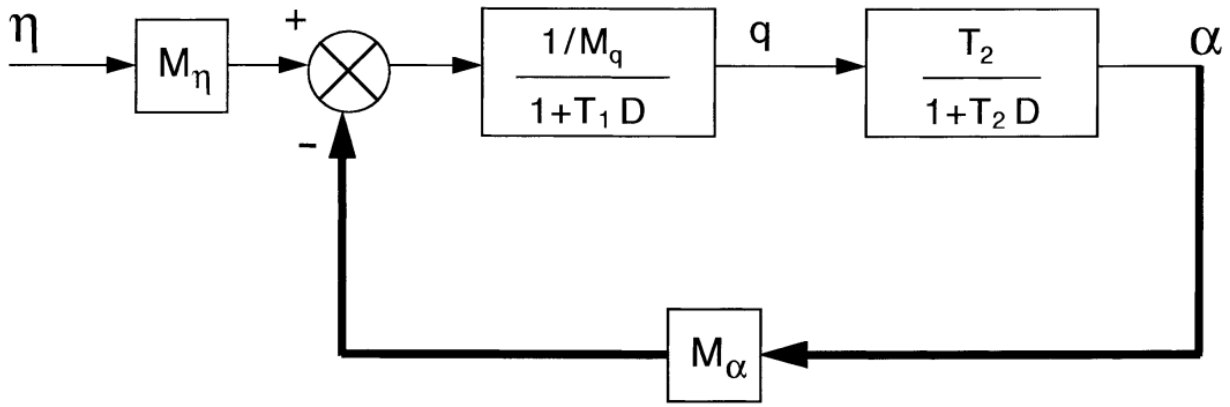


Fig. 3.30 Simplified overall block diagram.

Stability Augmentation System (SAS)

- SAS is the possible need for improved damping and stability about all three axes.
- Three-axis stability augmentation systems are installed in most high performance military jet aircraft and very many civil jet aircraft.

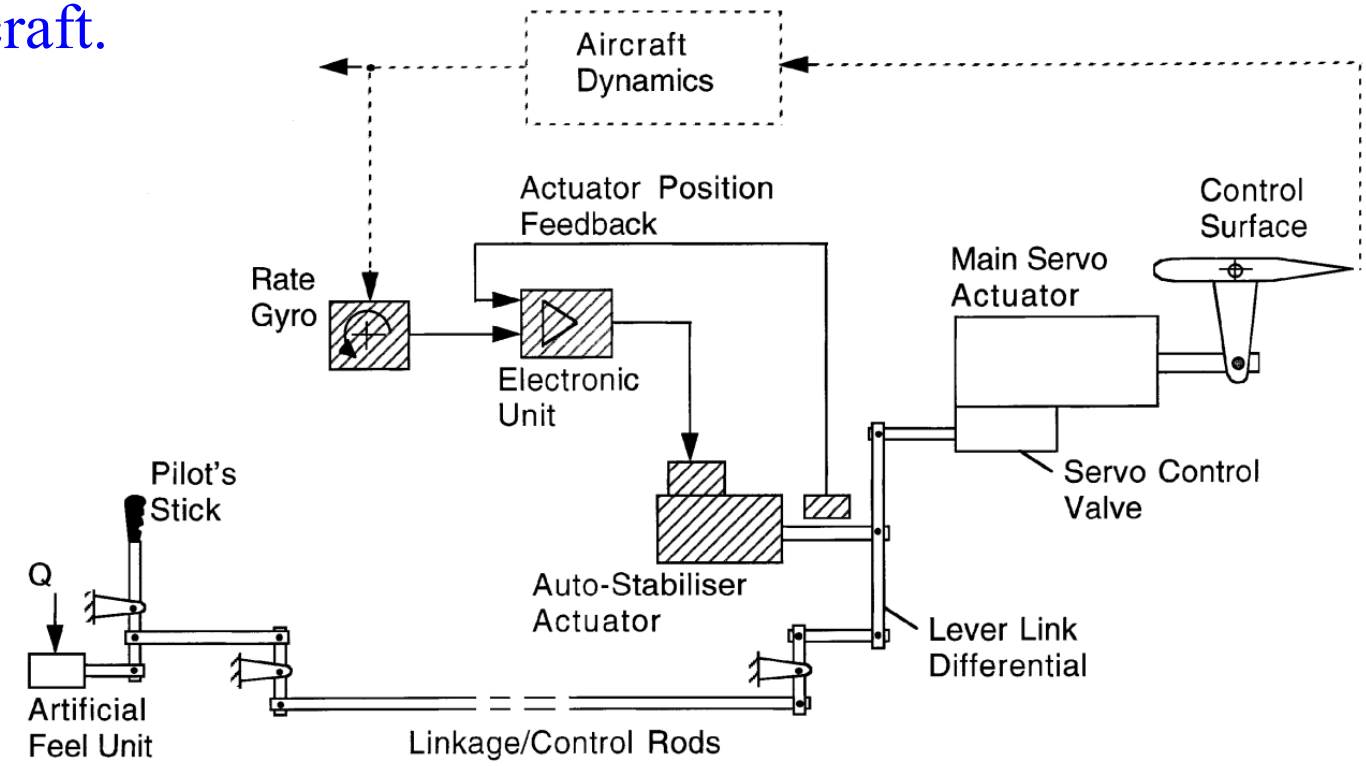


Fig. 3.37 Simple stability augmentation system.

Stability Augmentation System (SAS)

- Reduce the damping ratio of angular rates.

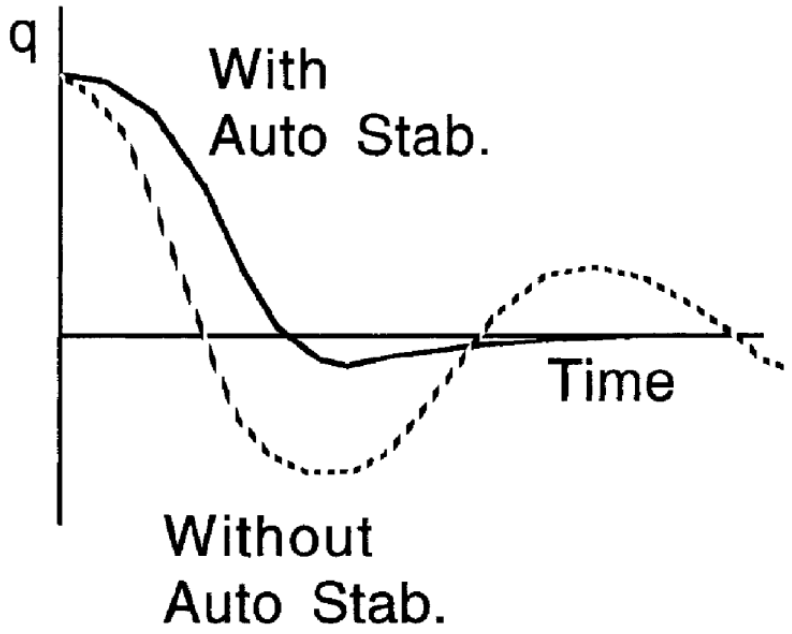


Fig. 3.40 Pitch response with and without stability augmentation.